WHO ACTUALLY CAPTURES THE INDIA–US MINERALS ALLIANCE?

Free Condensed Edition — Why Separation and Magnets Decide India's Place in the Hardware Century

Techadyant Labs

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About This Free Edition

This is a free, condensed edition of the Techadyant Labs strategic-intelligence report _**Who Actually Captures the India–US Minerals Alliance?**_ It carries the report's central argument, its analytical framework, and its headline findings — enough to be useful on its own. The complete 124-page edition — fourteen chapters with per-sector chokepoint scorecards, the quantified 2035 scenario model, the capital-stack analysis, the China-counterstrategy chapter, peer benchmarking, the policy roadmap, company-level beneficiary mapping, seven appendices and thirty figures — is available as a paid PDF.

Read or buy the full report: https://labs.techadyant.com/reports/who-actually-captures-the-india-us-minerals-alliance

Executive Summary

The thesis in one page

On 26 May 2026 India and the United States signed a critical-minerals framework. Both capitals read it as a supply deal — a way to move minerals out of the ground and away from China. That reading is wrong in a way that matters. The minerals at the centre of this story are not scarce in the earth's crust; India itself sits on roughly **7.23 million tonnes of rare-earth oxide** in monazite. They are scarce in *refined, separated, magnet-ready form* — and that capability is concentrated, deliberately and durably, in China.

The achievable and strategically decisive prize for India is therefore not more mining. It is to become the credible **third or fourth non-China node** in a handful of midstream capabilities — separated rare-earth oxides, sintered neodymium magnets, mature-node semiconductors, and the electronic-grade gases and chemicals that fabs consume. The alliance matters only insofar as it moves India up the two chokepoints that actually carry leverage: **separation and refining, and magnets and advanced materials** — not the two it talks about most, extraction and assembly, where India already has capability and where leverage is lowest.

The honest verdict is conditional optimism. The strategy is correct, the window is real, and the partner is motivated — the United States needs a non-China source of defence magnets badly enough to underwrite one. But India's success turns on the least glamorous and most failure-prone parts of industrial policy: building separation trains, clearing them environmentally, financing magnet lines through years of subscale economics, and aggregating enough demand to make any of it bankable. The headline numbers — lakh-crore fabs, million-tonne reserves — are not where this is won or lost. It is won or lost in the midstream.

The Four Chokepoints

The report's organising instrument is the **Techadyant Chokepoint Index**. Mineral value flows through four sequential chokepoints, and leverage concentrates in the middle two. India is scored 1–5 at each — against installed capability, funded trajectory, and sovereign control — and the same scorecard is applied to every sector, so semiconductors, electronics, defence, electric vehicles, energy and AI infrastructure are measured against one ruler.

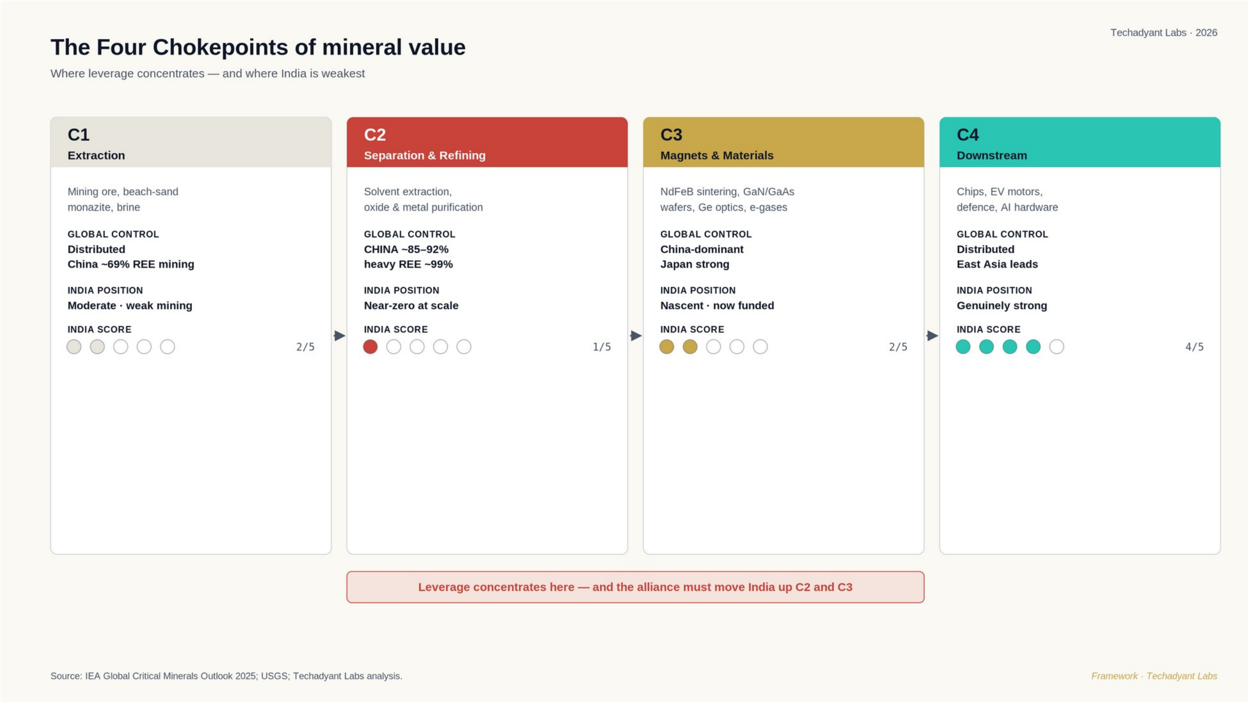

Figure 1 — The Techadyant Chokepoint Index: India is strong at the ends of its value chains and hollow in the middle.

The pattern that emerges is remarkably stable across sectors, and it is the report's central empirical finding: **India's weakness is not at the ends of its value chains but in their middles, and the alliance matters only insofar as it addresses the middle.**

Why processing, not reserves

China controls roughly two-thirds of rare-earth *mining* — a large share, but one that other countries are eroding. Its share of *separation and refining* is far higher, around nine-tenths, and approaches the entirety of commercial supply for the heavy rare earths that magnets require. It is this processing share, not the mining share, that confers leverage, because a separation plant — unlike a reserve — cannot be quickly relocated.

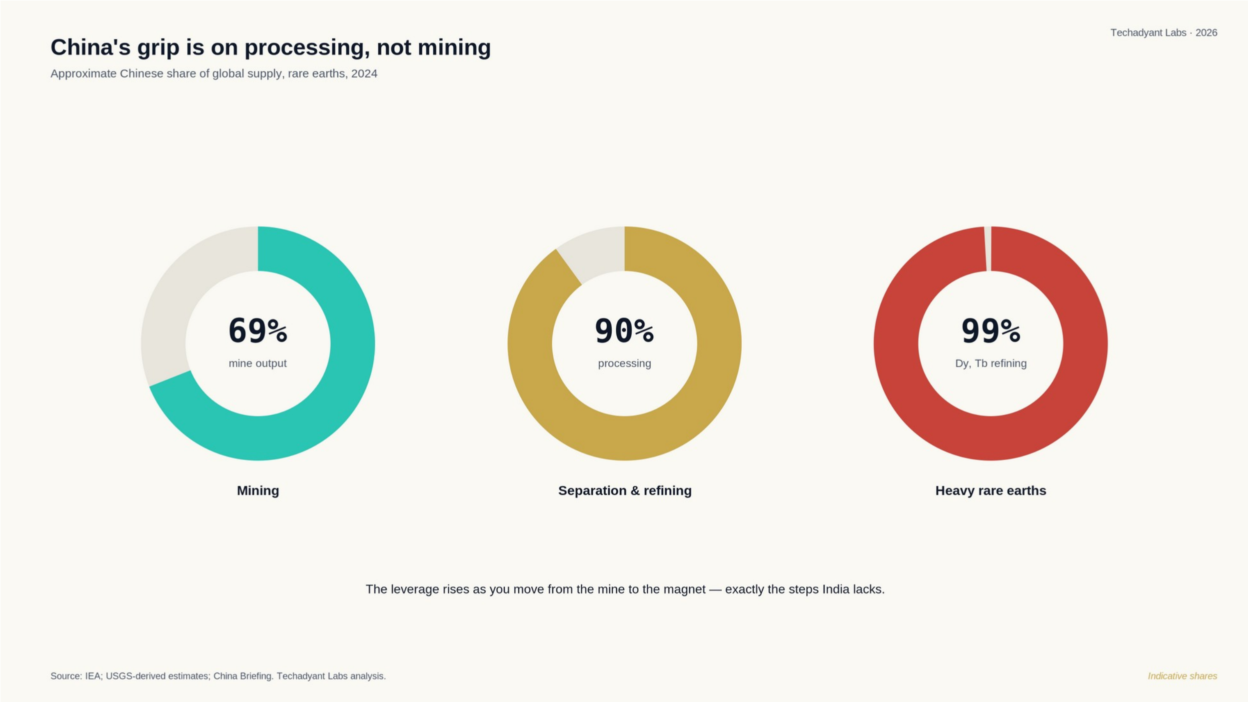

Figure 2 — China's share of global rare-earth supply rises sharply from mining to refining to heavy-rare-earth processing.

The 2024–25 export controls were the proof. When China restricted gallium, germanium and antimony, affected prices rose by several hundred per cent and export volumes collapsed toward zero. The leverage sat in the middle, where there is no fast alternative. The same flow — from mine to refined material to magnet — narrows to a Chinese chokepoint on which every downstream sector depends.

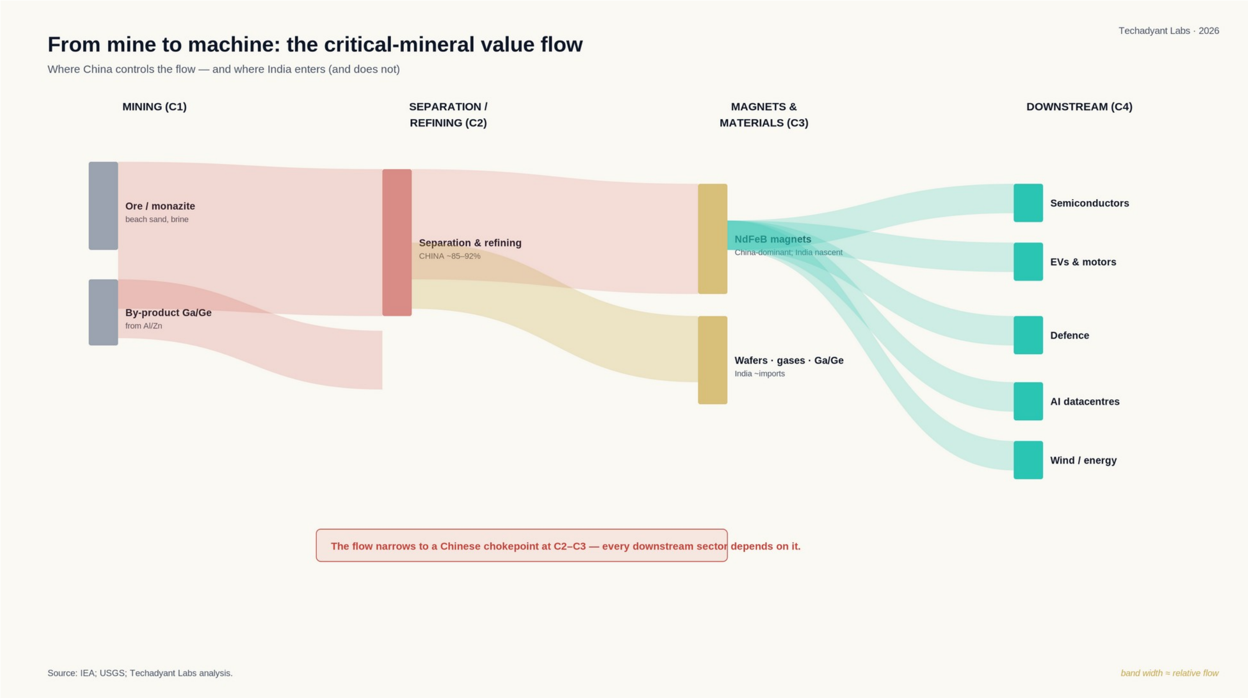

Figure 3 — From mine to machine: the value flow narrows to a Chinese chokepoint at separation and magnets.

The shared dependency

The diverse hardware ambitions India holds — chips, electric motors, guided weapons, batteries, wind turbines, AI servers — reduce to a small set of shared material dependencies. Because those dependencies are shared, the midstream that serves one sector serves all: it is not a sector-specific cost but shared industrial infrastructure. That is the strongest argument for building it once, at scale.

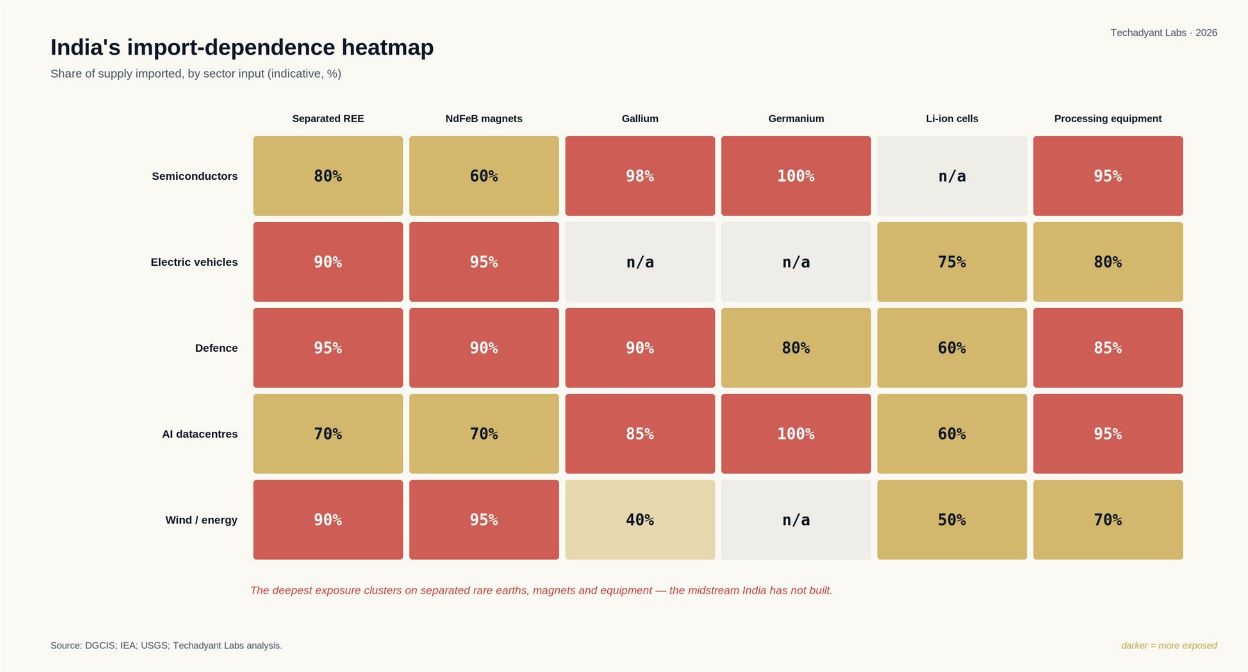

Figure 4 — India's import-dependence heatmap: the deepest exposure clusters on separated rare earths, magnets and equipment.

The same structural shape recurs everywhere: India is the world's second-largest assembler of mobile phones but captures only **18–20% domestic value**; it set a 50 GWh battery-cell target and built **1.4 GWh**; and it depends on China for roughly **99% of the heavy-rare-earth refining** that defence magnets require.

Who captures the value

The beneficiaries of the alliance, if it executes, are not the obvious ones. They are not primarily miners, and not the marquee chip and electric-vehicle brands. They are the layer in between: the **separation and refining operators** (IREL and the private entrants the schemes are designed to attract), the **magnet-makers** funded by the ₹7,280 crore permanent-magnet scheme, the **electronic-gas and specialty-chemical** producers any serious base requires, and the **recyclers** who can shortcut the separation problem. Downstream assemblers benefit too — but they gain *security of supply*, not *cost advantage*.

India against its peers

Benchmarked against the countries that matter — China, Japan, South Korea, Taiwan, Vietnam and Indonesia — India leads its emerging-economy peers on assembly and leads all peers except the East Asian incumbents on chip design. But on separation and refining it trails not only China but Japan, Korea and Taiwan. India's relative strength sits where leverage is lowest, and its relative weakness sits where leverage is highest.

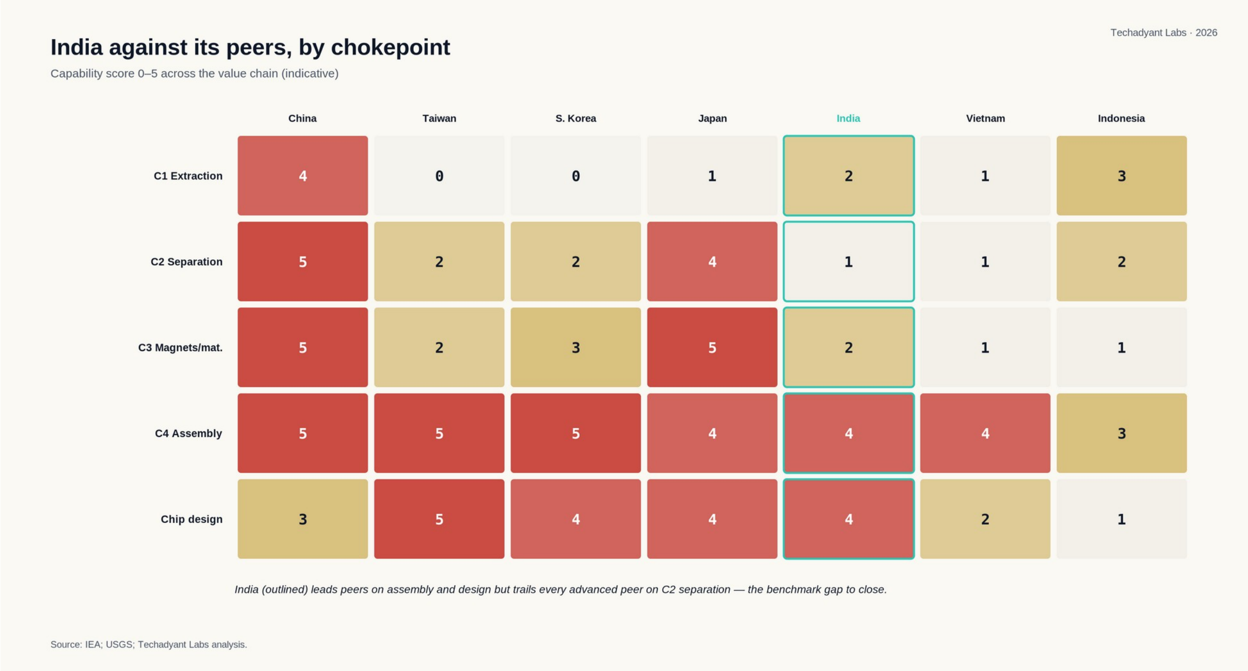

Figure 5 — India against its peers, scored by chokepoint: India leads on assembly and design but trails every advanced peer on separation.

Three scenarios to 2035

The report models three futures, distinguished by how far India moves up the second and third chokepoints. In the **conservative** case (highest probability), India builds a real but uneven midstream led by Gujarat and remains dependent on imported leading-edge chips and most precursors. In the **accelerated** case, magnet capacity reaches bloc-export scale, a second semiconductor cluster emerges, and battery cells finally cross from pilot to volume. In the **breakthrough** case (least likely), India becomes the global alternate for selected midstream products. The probability-weighted outcome is a credible fourth node, not a China substitute — but, measured against where India sits today, the largest industrial transformation since 1991.

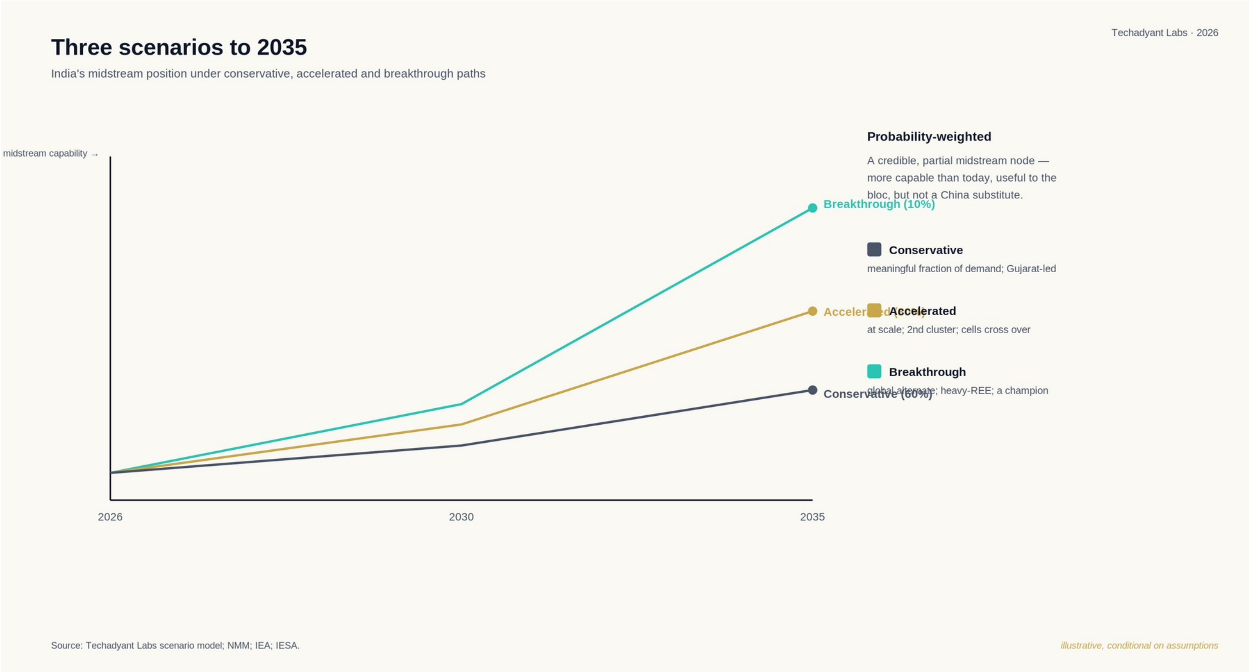

Figure 6 — Three scenarios to 2035: India's midstream capability under conservative, accelerated and breakthrough paths.

What this report recommends, in five sentences

Shift the weight of industrial policy decisively from the ends of the value chain — extraction and assembly, which have largely done their job — to the middle: separation, magnets, recycling and demand aggregation. Use the alliance to buy **technology transfer and guaranteed offtake**, not exploration acreage, because the scarce input is processing know-how and the United States' acute need for defence magnets gives India the leverage to demand it. Aggregate India's fragmented demand across sectors into a bankable signal, backed by floor prices, to neutralise the incumbent's ability to weaponise price. Redesign the failed battery-cell incentive on the principle of "fund the chain, not just the output." And build, starting now, the materials-and-process workforce — separation chemistry, metallurgy, magnet manufacturing — that the midstream cannot run without, because it takes years to train and the clock starts the moment the investment is committed.

Get the Complete Report

This free edition carries the argument. The complete 124-page edition carries it the way a decision-maker needs it:

- All fourteen chapters, each closing with a per-sector chokepoint scorecard
- The quantified 2035 scenario model with explicit assumptions
- The capital-stack analysis — who actually pays for the midstream, and the financing gap
- A dedicated chapter on China's likely counterstrategy and how India withstands it
- Peer benchmarking against China, Japan, Korea, Taiwan, Vietnam and Indonesia
- Company-level beneficiary mapping and the policy roadmap, phased to 2035
- Seven appendices — glossary, mineral profiles, semiconductor primer, policy timeline, company profiles, methodology, and the full layered bibliography
- Thirty figures, all on the Techadyant brand system

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